

Common to (CSE/IT/AIDS/AIML/CSE (CS)/CSE(IOT)) (Branches)

19AI301/CS3301 – Python Programming

PART A

2 Marks Questions

1. What will be the output of following code.

```
x = 3 / 2  
y = 3 // 2  
print(type(x), y)
```
2. What is the output of the following code?

```
a = 5  
b = 10  
a, b = b, a + b  
print(a, b)
```
3. List the literal collections in Python and give an example of each.
4. Difference between break and continue statement in python.
5. How to split strings and what function is used to perform that operation?
6. What will be the output?

```
d = {"a":1, "b":2}  
d["a"] = d.get("c", 0) + 5  
print(d)
```
7. Develop a program to count and print the number of occurrences of each character in the message using dictionary.
8. Distinguish between files and modules.
9. Write a Python program to create a new text file named example.txt and write some text into it.
10. What will be the output?

```
import numpy as np  
a = np.array([1,2,3])  
print(a + 2)
```
11. What is a Pandas Series? How is it different from a DataFrame?
12. Write a single Python statement to swap the values of two variables a and b without using a temporary variable.
13. Predict the output:

```
def test(n):  
    if n == 0:  
        return 1  
    return n * test(n-1)  
print(test(3))
```
14. Write a python program to add two numbers using anonymous function.
15. Predict the output:

```
L = [1, 2, 3]  
print(L * 2)
```

16. Given `t = (1, 2, 3, 2, 2)`, write a Python statement to count how many times 2 appears.
17. What is the role of the `__init__` method in a Python class? Provide an example to illustrate its usage.
18. Create a user-defined exception called `NegativeNumberError` and write a program to raise it if a negative number is input.
19. Given the following one-dimensional NumPy array:

```
import numpy as np
a = np.array([1, 2, 3, 4, 5, 6])
```

Reshape the above array into a 3-dimensional array of shape (1, 2, 3) and print the reshaped array along with its shape.
20. Create a Pandas Series with elements [10, 20, 30] and index labels ['a','b','c'].

PART B

13 Marks Questions

1. i) Write a python program to read two integers and perform simple arithmetic calculation.(+, -, *, /, //, %, **) (7 Marks)
ii) Develop a python program to get the details from the user such as employee name, age, address and designation, salary and display them. (6 marks)
2. Write short notes on types of operators in python with appropriate example.
3. i) Write a program that prints numbers from 1 to 20, skips multiples of 5 using continue, and stops if the number exceeds 15 using break.(7 Marks)
ii) Write a function `square_root(x)` that returns the square root of a number and call it for user input.(6 Marks)
4. i) Develop a Python program to count vowels and consonants in a string. (6 Marks)
ii) Write a program to gather individual's details such as name, age, and monthly income, and determine if the person is eligible for a housing loan. Eligibility criteria: age between 25 to 50 years and monthly income more than 30000. (7 Marks)
5. i) Write a Python program to implement Selection Sort. (7 Marks)
ii) Develop a Python program to create a histogram (frequency count) using dictionary. (6 Marks)
6. i) Write a program to remove all duplicate elements from a list. (6 Marks)
ii) Write a program to invert a dictionary (swap keys and values).
For example : `original_dict = {'a': 1, 'b': 2, 'c': 3}`
`inverted_dict = {1: 'a', 2: 'b', 3: 'c'}` (7 Marks)
7. i) Write a Python program to read a text file and count the number of words. (7 Marks)
ii) Develop a Python program to copy the contents of one file into another file. (6 Marks)
8. i) Write a Python program to create a class `Student` with data members and methods. (7 Marks)
ii) Develop a Python program to handle file not found exception using try-except block. (6 Marks)
9. Develop a Pandas program to split the following given DataFrame into groups based on school code and class. Also split the following DataFrame into groups based on class and create a list of names for each group. (13 Marks)
Original DataFrame:

	School code	class	name	date_Of_Birth	age
S1	s001	V	Alberto Franco	15/05/2002	12
S2	s002	V	Gino Mcneill	17/05/2002	12
S3	s003	VI	Ryan Parkes	16/02/1999	13
S4	s001	VI	Eesha Hinton	25/09/1998	13
S5	s002	V	Gino Mcneill	11/05/2002	14
S6	s004	VI	David Parkes	15/09/1997	12

10. i) Develop a python program to read all the information from the csv file 'nba.csv' and display the following
 1. First 5 rows. (2 Marks)
 2. Last 10 rows. (2 Marks)
 3. Find the number of rows in the file. (2 Marks)
 4. Find the number of columns in the file. (2 Marks)
 ii) Write a NumPy program to insert a new axis within a 2-D array. 2-D array of shape (5,3). New shape will be (5, 1, 3), (5, 3, 1), (1, 5, 3). (5 Marks)
11. Write the following python program.
 - i) To check whether a given number is Armstrong or not.(6 Marks)
 - ii) To print Fibonacci series. (7 Marks)
12. i) Develop a python program to get the details from the user such as student name, age, address and CGPA and display them. (7 Marks)
 - ii) Build a python program to get the principal, rate and time from the user and find the simple interest for it. (6 Marks)
13. i) Write a Python function to find the maximum of three numbers using recursive function. (4 Marks)
 - ii) Write a python function to calculate simple interest. Suppose the customer is a senior citizen, he is being offered a 10 percent rate of interest; for all other customers the rate of interest is 8 percent. (5 Marks)
 - iii) Write a python program to calculate the factorial of a number without recursion function. (4 Marks)
14. i) Write a program to determine if a **student passes or fails** based on marks in 3 subjects (pass if average ≥ 50). (6 Marks)
 - ii) Write a python program to check whether entered string is palindrome or not.(7Marks)
15. i) Write a python program to add two matrices using nested lists. (5 Marks)
 - ii) Write a python program to create a list of numbers in the range 1 to 10. Then delete all the even numbers from the list and print the list with odd numbers. (4 Marks)
 - iii) Write a python program to find the sum and mean of the elements in a list. (4 Marks)
16. i) Write a Python program to flatten a nested list using list comprehension.(5 Marks)
 - ii) Write a program to count the frequency of each character in a string using a dictionary. (5 Marks)
 - iii) Write a Python program to extract every second element from a list using slicing. (3 Marks)
 - i) Write a Python program to know the cursor position and print the text according to below-given specifications.
 1. Print the initial position. (2 Marks)
 2. Move the cursor to 4th position. And display next 5 characters. (2 Marks)

3. Move the cursor to the next 10 characters and print the current cursor position. (2 Marks)
4. Print next 10 characters from the current cursor position. (2 Marks)
- ii) Write a python program that uses class to store and display the employee's name and salary and print the total employee count whose details are available. (5 Marks)
17. i) Demonstrate using try, except, else, finally with file operations.(6 Marks)
- ii) Create a class Circle with a radius attribute and a method to calculate its area.(7Marks)
18. i) Write a Python function to find the maximum of three numbers using recursive function. (4 Marks)
- ii) Write a python function to calculate simple interest. Suppose the customer is a senior citizen, he is being offered a 10 percent rate of interest; for all other customers the rate of interest is 8 percent. (5 Marks)
- iii) Write a python program to calculate the factorial of a number without recursion function. (4 Marks)
19. i) Write a Pandas program to split the following dataframe into groups based on customer id and create a list of order date for each group. (6 Marks)

	ord_no	purch_amt	ord_date	customer_id	salesman_id
0	70001	150.50	2012-10-05	3005	5002
1	70009	270.65	2012-09-10	3001	5005
2	70002	65.26	2012-10-05	3002	5001
3	70004	110.50	2012-08-17	3009	5003
4	70007	948.50	2012-09-10	3005	5002
5	70005	2400.60	2012-07-27	3007	5001
6	70008	5760.00	2012-09-10	3002	5001

- ii) Write a Pandas program to join the two dataframes with matching records from both sides where available.(7 Marks)

data1:

	key1	key2	P	Q
0	K0	K0	P0	Q0
1	K0	K1	P1	Q1
2	K1	K0	P2	Q2
3	K2	K1	P3	Q3

data2:

	key1	key2	R	S
0	K0	K0	R0	S0
1	K1	K0	R1	S1
2	K1	K0	R2	S2
3	K2	K0	R3	S3

20. Develop a Pandas program to analyze employee data with the following specifications:
employee_data = {'name': ['Alice', 'Bob', 'Charlie', 'David', 'Emma', 'Frank', 'Grace', 'Hank', 'Ivy', 'Jack'],
'salary': [75000, 60000, 80000, np.nan, 55000, 95000, 72000, np.nan, 50000, 88000],
'daycount': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],

```
'eligibility': ['yes', 'no', 'yes', 'no', 'no', 'yes', 'yes', 'no', 'no', 'yes']}  
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

Perform the following task:

1. Select the rows where the daycount is less than 2 and salary is greater than 60000.
(5 Marks)
2. Display the names of employees whose salary is missing (NaN).(4 Marks)
3. Display the details of employees whose eligibility is 'no'. (4 Marks)

PART B

15 Marks Questions

1. A company tracks product sales in a **tuple list**: sales = [('Pen', 120), ('Notebook', 80), ('Pencil', 150), ('Eraser', 50)]. Complete the following task.
 1. Convert the tuple list to a dictionary (product → quantity). And Identify products with sales above 100. (7 Marks)
 2. Swap the highest and lowest selling product quantities. (4 Marks)
 3. Generate a text-based histogram of sales using asterisks. (4 Marks)
2. i) Write a python program to get 3 students' details such as student register no, name, department, blood group, English marks and store them as a list of dictionaries. Display the student name who scores the marks between 46 and 50. Add extra marks 3 and display the updated marks with the student's name. Also, display the students whose blood group is "B+" (8 Marks)
ii) Discuss any three python list methods with example programs. (7 Marks)
3. A company tracks product sales in a **tuple list**: sales = [('Pen', 120), ('Notebook', 80), ('Pencil', 160), ('Eraser', 60)]. Complete the following task.
 1. Convert the tuple list to a dictionary (product → quantity). And Identify products with sales above 100. (7 Marks)
 2. Swap the highest and lowest selling product quantities. (4 Marks)
 3. Generate a text-based histogram of sales using asterisks. (4 Marks)
4. i) Write a python program to get 3 students' details such as student register no, name, department, blood group, English marks and store them as a list of dictionaries. Display the student name who scores the marks between 46 and 50. Add extra marks 3 and display the updated marks with the student's name. Also, display the students whose blood group is "B+" (8 Marks)
ii) Discuss any three python list methods with example programs. (7 Marks)